

NPCI Advances India's Sovereign AI Infrastructure for Digital Payments with NVIDIA

The initiative builds on NPCI's AI journey, to enhance grievance redressal, safeguard data sovereignty, and unlock AI-driven innovation across India's digital payments ecosystem.

New Delhi, 18 February 2026: National Payments Corporation of India ([NPCI](#)) today announced its collaboration with NVIDIA to scale and advance its **sovereign AI model capabilities purpose-built for India's payments ecosystem**. The initiative will support the evolving requirements of large-scale, real-time payment systems, with an emphasis on trust, resilience, security, and ecosystem enablement.

The collaboration brings together NPCI's deep domain expertise in building and operating population-scale payments infrastructure with NVIDIA's advanced AI and accelerated computing platforms. As part of this engagement, NPCI will use [NVIDIA Nemotron](#) - a family of open models with open weights, training data and recipes - in its model development journey to create a payments-native AI foundation model aligned with India's regulatory and data sovereignty requirements.

NPCI's use of AI has been guided by practical requirements emerging from operating payment systems at scale. In this context, NPCI recently introduced the [UPI Help Assistant](#) as a pilot initiative, supported by [FiMI](#) (Financial Model for India) fine-tuned and pre-trained Small Language Model (SLM) developed specifically for the payment ecosystem. The assistant supports grievance resolution for UPI users by enabling more timely and consistent responses at scale.

As part of the next phase of its AI journey, NPCI aims to evolve from use-case-specific agents to a foundational, scalable AI layer for the payment ecosystem. The proposed model will explore architectures such as Mixture of Experts (MoE) to support high-volume, low-latency payment environments, while gradually expanding capabilities across multilingual datasets and agent-optimised systems.

Over time, the initiative is expected to support innovation across areas such as trust frameworks, grievance redressal processes, and operational intelligence. It is also intended to provide a platform that can be leveraged by banks, fintechs, and other participants in the payment ecosystem, while maintaining a strong focus on data security, sovereignty, and responsible use of AI.

Vishal Kanvaty, Chief Technology Officer, NPCI, said "Through this collaboration with NVIDIA, NPCI aims to advance AI capabilities designed specifically for India's payments ecosystem. Drawing from our experience of operating population-scale, real-time payment systems, this initiative is designed to create a sovereign, payments-native AI foundation that strengthens trust, resilience, and security, while remaining aligned with India's regulatory and data sovereignty requirements. As we evolve from use-case-led AI deployments to a foundational and scalable AI layer, our focus remains on enabling the broader ecosystem to innovate responsibly through robust governance frameworks and secure, future-ready infrastructure.

"India has one of the most advanced digital payment systems in the world that operates at population scale where trust, resilience, and performance are fundamental," said **Vishal Dhupar, Managing Director, Asia South, NVIDIA**. "With accelerated computing and AI, we aim to strengthen India's fintech infrastructure while enabling responsible innovation across the ecosystem," he added.

Through this collaboration, NPCI continues to focus on strengthening the underlying digital public infrastructure for payments, with a view to supporting long-term stability, efficiency, and innovation across the ecosystem.

About NPCI

National Payments Corporation of India (NPCI) is the central body responsible for overseeing retail payments and settlement systems in India. It is established by the Reserve Bank of India (RBI) and the Indian Banks' Association (IBA). The Company is focused on bringing innovations in the retail payment systems by using technology for achieving greater efficiency in operations and widening the reach of payment systems. NPCI is committed to harnessing the transformative potential of deep tech, creating an ecosystem that fosters collaboration to work on breakthrough technologies. Underlining its commitment to service, NPCI has been incorporated as a "Not for Profit" Company to provide infrastructure to the entire Banking ecosystem in India for physical as well as electronic payment and settlement systems.

NPCI has made a profound impact on India's retail payment landscape, focusing on creating robust, efficient, and inclusive payment and settlement solutions. NPCI has been instrumental in introducing a range of products that have revolutionised retail payment systems. These include [Unified Payments Interface \(UPI\)](#), [RuPay](#), [National Automated Clearing House \(NACH\)](#), [Immediate Payment Service \(IMPS\)](#), [National Electronic Toll Collection \(NETC\)](#), [Aadhaar Enabled Payment System \(AePS\)](#), [e-RUPI](#) and more. Each of these products has contributed significantly to enhance the efficiency and accessibility of payment systems in India, ultimately propelling financial inclusion.

NPCI has played a fundamental role in establishing the foundation for India's rapidly growing digital payments ecosystem, projecting the country onto the global stage.

NPCI has three wholly owned subsidiaries – [NPCI International Payments Limited \(NIPL\)](#) and [NPCI Bharat BillPay Limited \(NBBL\)](#) and [NPCI BHIM Services Limited \(NBSL\)](#), established in pursuance of NPCI Board & RBI approval.

For more information visit: <https://www.npci.org.in/>

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